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M1 MACRO

Group B

Session 3

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

Two period model with saving

$$\text{Max } U = \log C_1 + \beta \log C_2$$

Bx period 1: $C_1 + S = Y_1$

period 2: $C_2 = Y_2 + S(1+r)$

r is taken as given
= interest rate

Intertemporal Budget Constraint (IBC)

$$\begin{cases} C_1 + S = Y_1 \end{cases}$$

$$C_1 = Y_1 + (1+r)S \Rightarrow S = \frac{C_2 - Y_2}{1+r}$$

$$C_1 + \frac{C_2 - Y_2}{1+r} = Y_1 \quad \Leftrightarrow$$

$$\boxed{C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r}} \quad \text{IBC}$$

Opt. C_1 and C_2 of the household

$$\max \log C_1 + \beta \log C_2$$

st

$$C_1 + \frac{1}{1+r} C_2 = Y_1 + \frac{1}{1+r} Y_2$$

$$P_1 C_1 + P_2 C_2 = P_1 Y_1 + P_2 Y_2$$

$$\Leftrightarrow C_1 + \frac{P_2}{P_1} C_2 = Y_1 + \frac{P_2}{P_1} Y_2$$

in eq, $\frac{1}{1+r} = P_2 / P_1$

eq. price $P_1 = 1, P_2 = \beta \frac{Y_1}{Y_2}$

eq interest rate: $\frac{1}{1+r} = \beta \frac{Y_1}{Y_2}$

$$r = \frac{1}{\beta \frac{Y_2}{Y_1}} - 1$$

eq. interest
rate

check that $C_1 = Y_1$ for eq. r .

Stochastic endowment

y_1 given

$$y_2 = \begin{cases} \bar{y} + 1 = y_L & \text{prob } \frac{1}{2} \\ \bar{y} - 1 = y_H & \text{prob } \frac{1}{2} \end{cases}$$

$$E(y_2) = \bar{y}$$

• Agent's maximum **Expected utility**.

Consider the lottery: $C_2 = \begin{cases} C_{2L} & 1/2 \\ C_{2H} & 1/2 \end{cases}$

the utility of this lottery is

$$E u(C_2) = \frac{1}{2} u(C_{2L}) + \frac{1}{2} u(C_{2H})$$

[This is an assumption]

Problem of a household

$$\text{Max } \log C_1 + \beta E \log C_2$$

$$\hookrightarrow = \log C_1 + \beta \left[\frac{1}{2} \log C_{2L} + \frac{1}{2} \log C_{2H} \right]$$

BC $C_1 + S = Y_1$

$$\hookrightarrow C_2 = Y_2 + (1+r)S$$

$$\hookrightarrow \begin{cases} C_{2L} = Y_{2L} + (1+r)S \\ C_{2H} = Y_{2H} + (1+r)S \end{cases}$$

$$\text{Max } \log c_1 + \beta E \log c_2$$

$$\text{st } c_1 + \frac{c_2}{1+r} = y_1 + \frac{y_2}{1+r}$$

$$\text{Max } \log c_1 + \beta E \log \left[(1+r)(y_1 - c_1) + y_2 \right]$$

foc/

$$\frac{1}{c_1} = \beta(1+r) E \left[\frac{1}{c_2} \right]$$

$$C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r}$$

↳ means

$$\left. \begin{array}{l} C_1 + \frac{C_{2L}}{1+r} = Y_1 + \frac{Y_{2L}}{1+r} \\ C_1 + \frac{C_{2H}}{1+r} = Y_1 + \frac{Y_{2H}}{1+r} \end{array} \right\}$$

$$C_1 + \frac{C_{2H}}{1+r} = Y_1 + \frac{Y_{2H}}{1+r}$$